

MIDTERM MOCK TEST Case Study Practice

Course code: **CSW 309**

Course name: **Information Systems Analysis and Design**

Type of Exam: **Closed Book**

Exam code: **01**

Duration: **90 minutes**

Maximum mark: **100**

Instructions:

- *Non-programming calculator:* Yes ☒ | No ☐
 - *This paper consists of 2 sections and 5 practical questions. Section 1 contains 3 questions worth 40 marks in total. Question 1 carries 10 marks, Question 2 carries 10 marks, Question 3 carries 20 marks. Section 2 contains 2 questions worth 60 marks in total. Question 1 carries 30 marks, and Question 2 carries 30 marks.*
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Scenario: Appointment Scheduling for a Traditional Medicine Clinic

An **Lành Traditional Medicine Clinic** currently manages all patient appointments manually.

- **Current Process:** Patients call or visit the clinic directly to book an appointment. Staff members (Receptionists) record the Patient's information, the requested Service (treatment), and the appointment time in paper logbooks.
- **Problem:** This often leads to scheduling conflicts (especially when multiple Staff members are handling bookings) and consumes significant time searching for appointment details when a Patient arrives for their visit.
- **Requirement:** The management wants a **simple computerized system** that allows Staff to **record Patient information, manage the list of available services** (such as acupuncture, acupressure, and prescription), and **schedule appointments** for Patients. The system should also allow Staff to **view the list of appointments** for the day.

System Users:

- **Staff (Receptionist):** Manage Patient and Service information, Schedule appointments, View appointment reports.
- **Patient:** Request to book an appointment.

Main Processes:

- Manage Patient Information
- Manage Service Information
- Schedule Appointment
- View Appointment List

SECTION 1: Requirements & Use Case (40 points)

Question 1: Problem Description and System Request (10 pts)

- Summarize the **problem** (why the system is needed).
- Write the **Goal** and **2–3 expected benefits** of the new system.

Question 2: Functional Requirements (10 pts)

- List **5 main functions** the system must support.

Question 3: Use Case Diagram (20 pts)

- Draw a simple diagram showing **Staff** and **Patient** interacting with the system.
- Include the four main processes above as use cases.

SECTION 2: Process & Data Modeling (60 points)

Question 1: Data Flow Diagram (DFD) (30 pts)

- Draw one **Level-0 DFD**.
- Include at least **3 processes**, **2 data stores**, and **2 external entities**.
 - (e.g., *D1: Patient Data*, *D2: Service Data*)

Question 2: Entity–Relationship Diagram (ERD) (30 pts)

- Identify at least **3 entities** (e.g., Patient, Service, Appointment).
- Show attributes and relationships (PK/FK).

----- *End* -----